

Dhruv Khatri

New Delhi, India | dhruvkhatri1234@gmail.com | +91-981-889-6302 | github.com/dhruvicious
linkedin.com/in/bedhruvicious

TECHNICAL SKILLS

Languages	C++, JavaScript, TypeScript, Python
Backend & Databases	Node.js, Express.js, Quart, REST APIs, OAuth 2.0, JWT, PostgreSQL, MongoDB, SQLite
Frontend	React, Next.js, Tailwind CSS, Three.js, React-Three-Fiber
Tools & Platforms	Git, Docker, Linux, Gunicorn, Uvicorn, CMake, Prisma ORM, SQLAlchemy, Marshmallow
CS Fundamentals	Data Structures & Algorithms, System Design, Distributed Systems, OOP

ACHIEVEMENTS

- **Competitive Programming** — Solved 400+ problems on [LeetCode](#) and [Codeforces](#); attained **Specialist** rating (1400–1600) on Codeforces.
- **Hardware Hackathon (2023)** — 1st place; designed and built a functional Smart Mirror on Raspberry Pi within a 24-hour sprint.

PROJECTS

IoT Sensor Backend — Hub/Phone Fallback Architecture [dhruvicious/Major_Project](#)

Quart · SQLAlchemy async · aiosqlite · Docker · Gunicorn/Uvicorn · structlog

- Architected and rewrote a production-grade async REST API for IoT sensor node management, replacing a synchronous prototype with an end-to-end async stack (Quart + SQLAlchemy async + aiosqlite).
- Designed a hub/phone fallback system: hub heartbeat tracked per-request; on silence past a configurable threshold (default 60s), the API opens a phone fallback endpoint and switches data sources transparently.
- Implemented median-voting over rolling 5-reading windows on phone-submitted data for outlier rejection during hub outages.
- Deployed with Docker (multi-stage, non-root), Gunicorn + Uvicorn workers, structlog JSON logging, API-key auth, and `/healthz/readyz` probes.

BetterReads — Book Review Platform API [dhruvicious/betterReads](#)

Next.js · PostgreSQL · Prisma ORM · JWT · bcrypt.js

- Built a RESTful backend handling ~50k books and ~10k reviews under concurrent load, validated with synthetic data at scale.
- Secured all endpoints with JWT auth and bcrypt hashing; implemented role-based CRUD access control.
- Modeled relational PostgreSQL schema via Prisma ORM with indexed, high-concurrency queries and a clean, documented REST API surface.

expressCpp — Express.js-Style Web Framework in C++17 [dhruvicious/expressCpp](#)

C++17 · uWebSockets · CMake · Inja Template Engine · Catch2

- Designed and built a C++17 web framework porting Express.js's middleware/routing API onto a uWebSockets event loop: Radix-tree route dispatch, modular routers, a Connect-style middleware chain, and a thread pool offloading blocking file I/O.
- Implemented core HTTP primitives from scratch: request parsing, signed cookies, ETag generation, static file serving, and server-side rendering via Inja.
- Conducted a self-led post-mortem identifying concurrency, memory-safety, and security issues (race conditions, dangling-pointer risk, outdated hashing), documenting fixes for each.

CERTIFICATIONS

- **DSA in C++ & System Design** — Certificate of Excellence, Coding Blocks (2025).
- **Full-Stack Web Development (MERN)** — Certificate of Completion, Coding Blocks (2024).

EDUCATION

Guru Gobind Singh Indraprastha University, New Delhi	Sept 2022 – June 2026
B.Tech in Electronics and Communication Engineering	CGPA: 7.48/10